

FAQs – MMT-02

1. Pegging logic allows us to compete to always be first, and with discretion range, we take advantage of what we are willing to pay as soon as a suitable price comes in the order book? True or False?

Yes, true. With pegging orders, we can stay at whatever bid/ask level we want to be. And the discretion range allows us to be aggressive with prices when the opposite side order comes within our discretion range.

2. Does the excessive use of pegging and discretion logic not lead us to get a worse execution price?

Yes, it can lead us to get a fill at a worse price if we use discretion logic aggressively. As discussed in the class, we can have multiple execution algorithms that behave differently according to the problem statement and the priority of the execution.

3. How can we estimate the probability of order execution?

One can estimate the probability of the order execution from the historical tick by tick data (TBT) which is generally used by participants like hedge funds and prop desks. Such data includes events like order arrival in the order book, order modification, cancellation, execution, etc. One can subscribe to such data from the exchanges at a cost. Also, HFT players build their own order books from the tick-by-tick data they receive directly from the exchange. Using this data, we can estimate the probability of order execution at a particular level in the order book.

4. How do HFT players maintain their own order book when orders are executing in a fraction of a second?

HFT firms generally subscribe to true tick by tick data from an exchange. This subscription provides them with information on all events that happen on the exchange. The arrival of orders in the order book, orders being modified, orders getting cancelled, and execution of orders are examples of the events or ticks received in such data. A data packet over dedicated leased line connectivity (more on this in TIO-01 and TIO-02 lectures) transmits this information to subscribers. Such a setup and a state-of-the-art infrastructure allow them to build their own order books.

5. If we want to execute orders for delivery, do we need to worry about the bid-ask spread?

It depends on the size of the order that one wants to execute. If the order size is small, traders usually don't worry about the spread much. However, as the order size increases, one becomes wary of the spread, which can be significant.

6. If multiple users fire pegging orders, how would those orders interact? Won't that theoretically keep increasing the bid (or decreasing the ask)?

Let's say you implement pegging orders in your execution algorithm which pegs at the best bid. This means, whenever the best bid changes, your algorithm will modify the order and update the price to be the best bid. Now, let's say there is another trader who pegs at the best bid + 1 tick. Now, there can be a scenario where you keep pegging at the best bid, and the other user keeps pegging at best bid + 1 tick, and the asset price will keep

increasing unnecessarily. To avoid this, we need to be deliberate while pegging. There are several ways to avoid such a situation which we discussed in the class. For example, we can check the order size by which the other user is pegging or check how many traders are pegging ahead of us. Such sanity checks will help us avoid entering such loops.

7. If opposite side liquidity has iceberg orders, then what to do?

We as a trader usually don't have much control over the liquidity on the opposite side of the order book. If the opposite side order is the iceberg order, the order will keep refilling the liquidity until the whole iceberg order is filled.

8. For an iceberg order, if we divide the quantity into more partitions, then the transaction cost will increase. Why can't we execute orders in bigger size (or fewer chunks)?

Partitioning of an iceberg order largely depends on the opposite liquidity of the order book. If it is very thin, it doesn't make sense to fire orders with large sizes as it might sway the prices away. On the other hand, if there is enough liquidity on the opposite side, it might be spread across multiple levels of the order book and hence, might not give the desired price. As far as the transaction cost is concerned, it depends on how the broker charges. If the broker charges on the number of orders, then it will be a concern. However, if the broker charges depending on the quantity you trade, irrespective of the number of trades, the number of orders you fire would not matter.

9. If we place a limit order and then cancel it, how is it considered a fake order?

Doing it a few times unintentionally won't be considered spoofing. However, repeating this act many times with an intention to move prices is illegal in most of the markets and considered spoofing.

10. For an IOC order, do we get to input the time within which we want the order to be executed?

No, we don't get such an option. As the name suggests, such orders would be executed immediately (if there is enough liquidity on the opposite side), or it will be cancelled.

11. In situations where the bid-ask spread is very tight, how are pegging orders useful?

Pegging orders allow you to stay at the specified level in the order book no matter how wide or narrow the bid-ask spread. This implies that you'll have approximately uniform chances of order execution. Also, standing on a specific level in the order book ensures that you are not paying bid-ask spread if the opposite side market order hits your order.

12. Is it possible that after an order is modified, it will not get a fill? What happens in that case? Will the order revert to its previous state, or will it stay modified?

It is possible that an order may not get a fill even after it is modified depending on the market conditions. Once an order is modified, it will stay modified unless it is again updated manually. The modified order will remain in the order book until it is cancelled or executed.

13. Up to what price level can an order be pegged? Is there any predefined limit such that after several pegs, the order should be cancelled?

There are no such hard limits as up to how many times an order can be pegged. It depends on how we are developing the execution algorithm. We can build a logic such that an order pegs whenever the market price changes, or gets cancelled after pegging for a specific time.

14. What is the average time within which we should square-off the trade?

There is no such average time limit for squaring off the trades. It depends on the strategy's exit conditions and how frequently those conditions are met. By definition, the intraday strategies would have a small-time interval compared to strategies traded on a daily time frame.

15. Do exchanges provide the trader's name in the information/ticks when they broadcast ticks to their users?

Most of the exchanges do not provide information of a personal nature. All they provide is the information related to trades happening in real-time. Examples include new orders arrival, orders execution, modification, cancellation, etc.

16. Will the spread be more for low liquidity stocks? Do we require more pegging and discretion for such stocks?

Usually, the bid-ask spread will be higher for low liquidity stocks because of the thin volumes. Such stocks might not require more pegging but might demand a wide discretion range (because of the higher spread). Alternatively, we can use a blended approach of pegging and discretion based on the market conditions and dynamic discretion range.

17. With pegging or discretion range, more orders have to be fired up. Will that increase brokerage or the overhead costs?

Using pegging or discretion range might increase the number of orders which helps us in ensuring the fill. Accordingly, it might also increase the transaction cost. In Direct Market Access (DMA), the transaction cost isn't generally the issue.